

Assignment 0 — Statistical Foundations for Econometrics

Theme

A digital-finance company launched a financial-literacy intervention aimed at improving customer savings behavior.

The company collected customer-level data and now wants to understand:

- whether the intervention worked
- whether observed differences are statistically meaningful
- how uncertainty affects conclusions

This assignment is intended to introduce the following concepts:

- uncertainty
- sampling
- statistical inference
- variance and covariance
- hypothesis testing
- ANOVA and ANCOVA
- maximum likelihood estimation (MLE)

All analysis must be conducted using one realistic dataset.

Dataset

Use the **Student Performance Dataset** from UCI/Kaggle.

Original Variable	Interpretation
G3	Savings/performance outcome
studytime	Financial-literacy effort
higher	Treatment indicator
failures	Risk factor
absences	Behavioral variable
Medu, Fedu	Socioeconomic background
internet	Digital-access indicator
schoolsup	Intervention/support variable

Variable Mapping

You may rename variables if desired.

Main Research Question

Does educational/financial support improve student performance outcomes?

You should approach this as a simplified:

- policy evaluation problem
- behavioral intervention analysis
- financial-literacy study

PART A — Random Variables and Sampling

Tasks

Treat the following variables as random variables:

- G3
- studytime
- absences
- failures

Sampling Exercise

Take 100 random samples of sizes:

- $n = 20$
- $n = 50$
- $n = 100$

For each sample:

- compute the sample mean of $G3$
- store all estimates

Create:

- histograms of sample means
- sampling-distribution visualizations

Required Output

Tables

Sample Size	Mean of Sample Means	Std. Dev. of Sample Means
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Plots

- Histogram of sample means
- Sampling distribution visualization

Questions

Explain:

- Why do estimates vary across samples?
- Why does larger n reduce uncertainty?
- What is sampling noise?

PART B — Estimators, Bias, and Variance

Tasks

Estimate:

- sample mean
- sample variance
- group averages

for:

- G3
- studytime
- absences

Simulation

Compare:

- small samples
- large samples

Show how the variance of the estimator changes.

Required Output

Sample Size	Variance of Mean Estimator
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Questions

Discuss:

- bias
- variance
- bias-variance tradeoff
- unbiased estimators

PART C — Variance, Covariance, and Correlation

Tasks

Compute:

- variance
- standard deviation
- covariance matrix
- correlation matrix

for:

- G3
- studytime
- failures
- absences
- Medu
- Fedu

Required Output

Heatmaps

- Correlation heatmap
- Covariance heatmap

Scatter Plots

- studytime vs G3
- absences vs G3
- failures vs G3

Questions

Explain:

- difference between covariance and correlation
- positive vs negative relationships
- limitations of correlation

PART D — Confidence Intervals and Standard Errors

Tasks

Construct:

- 95% confidence interval for mean G3
- 95% confidence interval for mean G3 by gender
- 95% confidence interval for studytime groups

Compute:

- standard errors
- margin of error

Required Output

Group	Mean	Std. Error	95% CI
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Questions

Explain:

- interpretation of confidence intervals
- effect of larger sample sizes
- difference between standard deviation and standard error

PART E — Hypothesis Testing

Tasks

1. Difference in Means

Test whether students with school support ($schoolsup = 1$) have higher G3.

Use:

- two-sample t-test

2. Internet Access

Test whether students with internet access perform better.

Required Output

Hypothesis	t-statistic	p-value	Reject H_0 ?
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Questions

Explain:

- null hypothesis
- alternative hypothesis
- Type I error
- Type II error
- practical vs statistical significance

PART F — Correlation vs Causation

Tasks

Analyze relationships between:

- studytime and grades
- internet access and grades
- school support and grades

Questions

Discuss:

- confounding variables
- reverse causality
- selection bias
- why correlation does not imply causation

Provide one misleading-correlation example.

Example:

Students receiving school support may already be weaker students.

PART G — ANOVA

Tasks

Test whether mean grades differ across:

- studytime categories
- parent education groups

Use:

- one-way ANOVA

Source	Sum Sq	df	F-statistic	p-value
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Required Output

Questions

Explain:

- between-group variation
- within-group variation
- interpretation of the F-statistic

PART H — ANCOVA

Tasks

Estimate the effect of:

- school support

on:

- G3

while controlling for:

- studytime
- absences
- failures
- parent education

Model:

$$G3 = \beta_0 + \beta_1 \text{schoolsup} + \text{controls} + u$$

Required Output

Variable	Coefficient	Std. Error	p-value
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Questions

Explain:

- role of controls
- difference between ANOVA and ANCOVA
- reduction in unexplained variance

PART I — Maximum Likelihood Intuition

Tasks

Assume:

- $G3$ follows a normal distribution

Estimate:

- mean
- variance

using MLE intuition.

Use:

```
scipy.stats.norm.fit()
```

Required Output

- Histogram of $G3$
- Fitted normal distribution curve

Questions

Explain:

- what likelihood means
- difference between probability and likelihood
- why MLE chooses parameters maximizing observed-data likelihood

Final Deliverables

Submit the following:

1. Jupyter notebook
2. Short report (4–6 pages)
3. Statistical tables
4. Visualizations
5. Interpretation of all tests and results

Main Learning Goal

By the end of this assignment, you should understand:

- how uncertainty appears in data
- how estimators behave
- how sampling affects conclusions
- how inference works
- how hypothesis testing works
- the difference between correlation and causation
- how variance and covariance affect analysis
- basic statistical reasoning for econometrics